

## Topology PhD Exam, May 2008

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*Work the following problems and show all work. Support all statements to the best of your ability.*

1. Compute  $\pi_1(S^2)$  and  $H_*(S^2)$ .
2. Let  $X$  be a connected completely regular topological space having more than one point. Can  $X$  be countable?
3. Does there exist a covering space of the figure eight with nontrivial abelian fundamental group?
4. Show that the 2-dimensional sphere with four points deleted cannot be a topological group.
5. Prove that there is no map of degree two from  $S^2$  to the torus  $T^2$ .

*Answer the following with complete definitions or statements or short proofs.*

6. State the Baire Category Theorem.
7. State the homology Mayer–Vietoris Theorem.
8. State the Urysohn Lemma.
9. State the Five Lemma.
10. Compute the Euler characteristic  $\chi(\mathbb{R}P^4 \times S^3 \times T^2)$ .
11. Define retraction and deformation retraction.
12. State the Lefschetz Fixed Point Theorem.
13. Can irrational numbers be presented as a countable union of closed in  $\mathbb{R}$  subsets?
14. Draw a picture of the universal cover of the 2-sphere with the segment joining the north and south poles.
15. Describe all connected subsets of  $\mathbb{R}$ .